



Handout 7.7: Problem Scan, Idea Filter, and Role Cards

Unit 7, Lesson 7.7. **Print pages 1 and 2 one per student. Print page 3, the role cards, one per team of three, on card stock if you have it.**

Team members: _____

Date: _____ Period: _____

Page 1: Problem scan

Six signs are posted around the room: FOOD, MONEY, FAMILY, HOME, CLOTHING, and WORK. Visit three of them. Under each one, write problems you have **actually seen**. Not problems you imagine. Drawings count. One word counts.

Sentence frame, English and Spanish

English	Spanish
The problem is . has this problem.	El problema es . tiene este problema.

Sign 1: _____ (FOOD, MONEY, FAMILY, HOME, CLOTHING, or WORK)

Problem I have seen	Who has it

Sign 2: _____

Problem I have seen	Who has it

Sign 3: _____ (one of your three stops must be WORK)

Problem I have seen	Who has it

Count your team's problems: _____ (target: at least 8 per person, 24 or more per team)

Page 2: The idea filter

Step 1: Twenty ideas. No judging.

Six minutes. One idea per sticky note. Look at each problem on page 1. What could you make or do that would solve it? **Nobody says “that’s dumb.” If you want to say it, say “later” instead.**

Our team’s idea count: _____ (target 20)

Step 2: Run every idea through three questions

Ask the three questions about every idea. If an idea fails **any one** of the three, cross it out. Do not argue about it. Cross it out.

	The question	Why it matters
1	Is the problem real? Has somebody in this room actually had it?	If nobody has the problem, nobody pays.
2	Could three middle schoolers actually build or do this in five days with what is in this room?	You pitch on Friday. Not next year.
3	Would anyone pay for it, and about how much?	Free is not a business.

Write your surviving ideas here.

Idea	Q1 real?	Q2 buildable in 5 days?	Q3 would pay? About how much?

Step 3: Rank the survivors

Rank	Idea
1	

2	
3	

Grade 8 line, fourth filter question: Who is already solving this problem, and what would make somebody choose you instead?

Already solving it: _____

Why somebody would pick us: _____

Step 4: The gate. Hand this in before you leave.

Our idea: _____

It is a product / service (circle one)

The FACS module it connects to (circle at least one):

FOOD (Units 1 and 6) | MONEY (Unit 2) | FAMILY (Unit 3) | HOME (Unit 4) | CLOTHING (Unit 5) | WORK (Unit 7)

Our one-sentence reason:

Our idea connects to _____ because _____

Line 2: _____

Sentence frame, English and Spanish

English	Spanish
Our idea connects to ___ because ___.	Nuestra idea se conecta con ___ porque ___.

What we will put in a judge's hands on Friday (one sentence, be specific):

Line 1: _____

Our roles (see the role cards on page 3):

CEO: _____ CFO: _____ CMO: _____

Signatures (all three):



Teacher initials (the gate): _____

Page 3: Role cards

Cut these three apart. One per team member, kept in a pocket or taped inside the team folder.

Read this line out loud before you pick: nobody is the boss of the other two. Every role does real work, and every role's name gets said out loud on Friday.

CEO (Chief Executive Officer)

Name: _____

You are in charge of the idea, the pitch, and the clock.

Your jobs:

- Write the **business brief**: the problem, who has it, the product or service, one thing that makes it different, and the FACS module sentence.
- Write **panel 5, the ask** (what you are asking the judges for).
- Say the **hook** and the **ask** in the pitch.
- **Call the time**. Twice a period, out loud: "We have __ minutes left."
- When the team is arguing instead of working, say so and pick.
- On Friday: take the first judge question unless the team picked somebody else.

On Friday a judge may ask you: "What is your margin?" (Margin is the price minus what one costs you to make.) You have to know it without looking at the poster.

You are the right CEO if: you can keep three people moving without being mean about it. And you can make a decision when two people disagree.



CFO (Chief Financial Officer)

Name: _____

You are in charge of every number.

Your jobs:

- Build the **numbers page**: unit cost line by line, price, margin per unit, fixed cost, break-even in units, and the one-month projection. (You learn every one of these in Lesson 7.8.)
- **Sign the numbers page**. Your signature means the arithmetic is right.
- Draw **panel 4** so the numbers can be read from ten feet away.
- Say the **money sentence** in the pitch.
- On Friday: answer “what does one cost you and what do you charge” without hesitating.

Your sentence to memorize: “We stop losing money after ___ units.”

You are the right CFO if: you like numbers and you are careful. And you are willing to be the one who says “that does not add up.”



CMO (Chief Marketing Officer)

Name: _____

You are in charge of the customer, the look, and the prototype (the sample you hand a judge).

Your jobs:

- Run the **three customer interviews** and write the answers in the customer's **own words**, including one quote each.
- Write **panels 1, 2, and 3**: the problem, the solution, the customer.
- Lay out all five panels of the poster so somebody reads them in the right order.
- **Build the prototype** or lead the building of it, using a skill from this year.
- Say the **problem** and **who it is for** in the pitch.
- On Friday: name one real person who said they would buy it, and what they said.

You are the right CMO if: you can talk to people you do not know. Or you can draw or lay something out. Or you built something good this year.

If you are better at drawing or building than at writing, CMO is your role. Say so out loud.

All three of you

- Know **all four numbers**: unit cost, price, margin, break-even.
- **Speak part of the pitch**. How much is up to you.
- Be able to say the **FACS module sentence** from memory.

How to pick, if nobody wants to decide

Answer these three questions out loud as a team. Do not vote. Talk.

1. **Who kept the team on time during the Unit 6 cook day, or during any group project this year?** That person is a strong CEO.
2. **Who did the arithmetic when the team had to cost something?** That person is a strong CFO.
3. **Who built, drew, sewed, or laid out the thing that came out best this year?** That person is a strong CMO.

If two people want the same role, the one who has **not** had that kind of job before takes it. You have one week left in this course. This is the last chance to do a job you have never done.